

Introduction to Artificial Intelligence

What does it mean to be intelligent?

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What is AI?

What do you think is Artificial Intelligence?

- Clockwork devices / Automaton
- Calculators
- GPS navigation
- Vending machines
- Robots
- Chess engines
- Expert systems
- Google search
- Spam filters
- Image recognition software
- Self-driving cars
- ChatGPT

What is AI?

Usually considered abilities related to Intelligence:

- Knowledge
- Reasoning and problem-solving
- Learning
- Communication

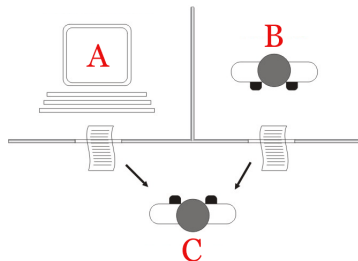
In AI, two main dimensions are considered:

- Think vs Act
- Rationally vs Like a human

Think rationally	Think like humans
Act rationally	Act like humans

Act like humans: Turing Test

Proposed by Alan Turing in 1950.



Criticisms:

- Behavioural: only testing the capacity of imitate human-like behaviour
- Not reproducible: results depend on the interrogator
- Anthropocentric: assumes human intelligence is the 'standard'

Act like humans: Turing Test

What does the computer need to do to pass the Turing Test?

- Natural Language Processing
- Knowledge Representation
- Automated Reasoning
- Machine Learning

Total Turing Test

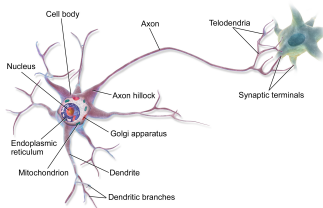
- Computer vision
- Robotics

Think like humans: Cognitive Science

Interdisciplinary study of how the mind works including:

- reasoning
- learning
- perception
- memory
- psychology
- neuroscience

Nowadays, cognitive science/neuroscience are used to inspire AI models (e.g., neural networks), but are considered separate fields.



Think rationally: Logic

The idea of using formal logic to produce rational behavior

- Syllogism: “All men are mortal. Socrates is a man. Therefore, Socrates is mortal” (Aristotle, 384-322 BC).
- The Laws of Thought (Boole, 1854)
- Hilbert’s 2nd Problem (1900), and Gödel incompleteness (1931).
- Expert systems (1960s-80s)

Limitations:

- Some aspects of human thinking are difficult to capture in formal logic (e.g., common-sense reasoning).
- Computationally expensive (because exact reasoning is hard).
- Not everything can be deduced.

[More in the Knowledge Representation course (2nd year).]

Think rationally: Probability

Logic produce only “exact” reasoning, but human thinking also needs to reason under “uncertainty”.

- Probability theory (Bayes, 1763)
- Bayesian networks (Pearl, 1985)
- Markov decision processes (Bellman, 1957)

Acting rationally: Rational Agent

Agent: an entity that perceives its environment and acts consequently.

Important:

- “rational” means aim to maximize performance measure;
- maximize performance is wrt what the agent knows;
- rational does not necessary require to think!

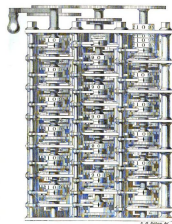
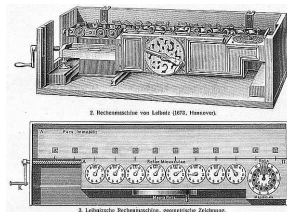
History

Automatons [act like humans]

- From Pindar's seventh Olympic Ode (5th century BC):
*"The animated figures stand
Adorning every public street
And seem to breathe in stone,
or move their marble feet."*
- Jacques de Vaucanson (1738)
- Frankenstein by Mary Shelley (1818)
- Yumihiki-Doji (1854) or Floutiste (1869-77)
- The term Robot first appears in the 1921 play 'R.U.R.' (Rossum's Universal Robots) by the Czech writer Karel Čapek.
 - Maria in Metropolis (1927)
 - Isaac Asimov (1942)
 - H.A.L. 9000 in 2001: A Space Odyssey (1968)
 - C-3PO and R2-D2 in Star Wars (1977)
 - The Terminator (1984)

Machines that compute ["think" rationally]

- Abacus (circa 2400 BC)
- Antikythera mechanism (circa 100 BC)
- Leibniz's calculating machine (1673-94)
- Babbage's Difference Engine (1822) and Analytical Engine (1837)
- Ada Lovelace's notes on the Analytical Engine (1843)
- Turing Machine (1936)
- Von Neumann architecture (1945)



1940s–1950s: Early Foundations

- Alan Turing proposes the “Turing test” (1950)
- Warren McCulloch and Walter Pitts develop the first neural network model (1943)
- Claude Shannon applies information theory to chess (1950)
- John McCarthy uses the term "Artificial Intelligence" (1955)
“the science and engineering of making intelligent machines”

History of AI (cont.)

1960s–1980s: Growth & Challenges

- Progress in symbolic AI and natural language processing
- Introduction of neural networks (Perceptron)
- "AI winters" due to unmet expectations (limits of computability)
- Rule-based systems for domain-specific problems (Expert systems)

1980s–1990s: Revival and Hybrid Approaches

- Back-propagation algorithm revitalizes neural networks (1986)
- Integration of symbolic and sub-symbolic methods
- Increased computational power enables more complex models
- IBM's Deep Blue defeats Garry Kasparov (1997)

History of AI (cont.)

2000-present: Big Data and Deep Learning

- Explosion of data and computational resources (WWW)
- Deep learning breakthroughs in image and speech recognition
- AI applications in various industries (healthcare, finance, etc.)

2020s–Present: AI Everywhere

- AI in healthcare, finance, robotics, and more
- Generative AI (e.g., ChatGPT)
- Focus on ethics, fairness, explainability

Some AI Paradigms

Symbolic AI (Good Old-Fashioned AI - GOFAI)

- Strengths: Transparent reasoning, excels in structured tasks
- Weaknesses: Struggles with perception, requires manual knowledge engineering
- Foundation: Logic, rules, symbolic representations

Statistical AI (Machine Learning)

- Strengths: Learns from data, handles uncertainty well
- Weaknesses: Requires large datasets, less interpretable

Key Challenges in Achieving AGI

- Complexity of Human Intelligence
- Common-Sense Reasoning
- Transfer Learning
- Ethical and Safety Concerns
- Resource Requirements
- Mathematical limitations

Digital Computers and AI

- The digital general-purpose computer opens up new opportunities.
- Develop software that performs complex symbolic operations, leading to new methods and advances in Science.
 - Problem-solving (Checkers, Chess)
 - Expert systems (medical diagnosis, geological analysis)
 - Theorem-proving (general problem solver)
 - Simulations (chemistry, biology, etc)
 - Pattern recognition (image recognition, speech recognition)
 - Machine learning (data analysis, predictions)
 - Natural language processing (translation, chatbots)
 - Quantum computing

Summary

- The idea of building intelligent machines dates back to ancient times.
- Digital computers allow processing complex tasks.
- AI is wide: formalization, problem-solving, learning, ...
- Different approaches to AI: symbolic, statistical, neural networks, ...

Book references:

- Chapter 1 and Section 28.1-2

Next lecture:

- Rational Agents